6D Object Pose Estimation for Robotics: Trading Generalization, Real-Time Capability, and Accuracy

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Abstract— **Robotic manipulation depends on localizing objects in six degrees of freedom (6D), three of translation and three of rotation. This report analyzes the development of 6D object pose estimation from classical geometric registration and template matching through supervised deep learning and dense correspondence learning to zero-shot methods built on vision foundation models. The literature is evaluated against the three requirements of robotic deployment, namely generalization to novel objects, real-time capability, and accuracy. The published evidence shows that no single current method satisfies all three. Frozen foundation features have largely closed the accuracy gap between training-free and object-specifically trained methods, but have per-image runtimes one to three orders of magnitude above the 33 ms frame budget of camera-rate perception. Because the published runtime profiles locate the cost in the global pose search rather than in the per-frame update, we identify decoupled hybrid architectures combining slow zero-shot initialization with fast local tracking as the most viable solution toward satisfying all three requirements, and outline the problems such architectures leave open.**

I. INTRODUCTION

An autonomous robot that grasps a tool, inserts a component, or hands an object to a human must first determine where that object is. Detection in the image plane does not suffice. Grasp and motion planning require the object's full 6D pose, i.e., its 3D position and 3D orientation relative to the camera [1]. The demand spans industrial bin picking, where the part set changes with every product cycle, and household robotics, where the object set is open-ended. In both settings, the perception system must deliver poses accurate enough for manipulation, fast enough for closed-loop control [2], and available for objects it has never seen before.

The field has produced diverse approaches to this problem in the past five decades, as illustrated in Fig. 1. Abdel-Aziz and Karara [3] first recovered the pose in closed form with the Direct Linear Transformation (DLT), solving a linear system from point correspondences assumed already known, a formulation later given a numerically stable solution by Fiore [4]. Rusu *et al.* [5] instead established such correspondences directly on depth data, encoding local surface geometry as histograms of relative normal orientations, known as Fast Point Feature Histograms. Hinterstoisser *et al.* [6] sidestepped explicit correspondences altogether in LINEMOD, matching the scene against precaptured per-viewpoint templates that fuse color gradients and surface normals. With the advent of deep learning, Peng *et al.* [7] and Wang *et al.* [8] replaced hand-crafted features with representations learned end-to-end for a fixed object set, voting for keypoints in the PVNet pipeline or fusing appearance and geometry in DenseFusion to predict the pose. Haugaard and Buch [9] pushed this idea to its most expressive form with SurfEmb, learning a continuous distribution over dense 2D–3D correspondences that absorbs object symmetries without manual annotation. Most recently, Wen *et al.* [10] in FoundationPose (FP) and Caraffa *et al.* [11] in FreeZe reused general-purpose visual and geometric features from vision foundation models to localize novel objects with no per-object training.

We argue that no single current method simultaneously satisfies the three requirements needed for robotic manipulation. These are generalization to novel objects without per-object training, denoted as R1, real-time capability compatible with closed-loop control, denoted as R2, and state-of-the-art accuracy, denoted as R3. Frozen vision foundation features have recently removed the conflict between R1 and R3, as training-free methods now match object-specifically trained pipelines on unseen objects [12], but their runtimes remain far above control-loop budgets.

We therefore argue that architectures that decouple slow initialization from fast tracking currently offer the best path toward satisfying all three requirements jointly.

The scope of this report is instance-level pose estimation of rigid objects from a single RGB, RGB-D view or point cloud. While category-level pose estimation also aims at generalization, it focuses on unseen instances of known categories without requiring CAD models at test time. This report instead focuses on zero-shot instance-level generalization, or R1, where arbitrary novel objects must be localized given their CAD models or reference views. Multi-view fusion and the upstream 2D detection literature are covered only where they intersect this analysis. The analyzed methods are chosen as the paradigm-defining or best-performing representatives of each phase in Fig. 1, following the BOP Challenge 2024 results [12] for the zero-shot era.

The report is organized as follows. Section II formalizes the pose estimation problem, its evaluation metrics, and the deployment requirements. Section III analyzes classical geometric approaches, focusing on Fast Point Feature Histograms (FPFH) [5] and the multi-modality tracker ICG+ [13]. Section IV traces the learning-based development from supervised, object-specific networks through dense correspondence learning in SurfEmb [9] to zero-shot foundation-model methods such as FP [10] and FreeZe [11].

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Section V synthesizes the evidence into a trade-off analysis and discusses hybrid architectures, open problems, and development perspectives, before Section VI concludes.

II. THE 6D OBJECT POSE ESTIMATION PROBLEM

A. Problem Formulation

Given sensor data of a scene in the form of an RGB image, a depth map, or both and a representation of a rigid target object as a CAD model, reference images, or a learned model, 6D pose estimation seeks the rigid transformation from the object frame to the camera frame,

$$\mathbf{T} = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0} & 1 \end{bmatrix} \in SE(3), \tag{1}$$

where $\mathbf{R} \in SO(3)$ is a rotation matrix and $\mathbf{t} \in \mathbb{R}^3$ a translation vector.

This can then be used for grasp and motion planning requiring the pose in the robot base frame B, obtained through the transformation chain,

$$\mathbf{T}_O^B = \mathbf{T}_C^B \, \mathbf{T}_O^C, \tag{2}$$

where $\mathbf{T}_X^Y$ denotes the transformation from frame X to frame Y and $\mathbf{T}_O^C$ is the estimate produced by the perception system.

For a fixed, externally mounted camera in the eye-to-hand configuration, $\mathbf{T}_C^B$ is static and determined once by offline hand-eye calibration, while for a wrist-mounted camera in the eye-in-hand configuration, it changes with every joint motion and follows from forward kinematics and the camera-to-flange calibration. In either configuration, the accuracy of the final grasp is bounded by the accuracy of the perception estimate, which motivates the accuracy requirement introduced below.

Two problem variants must be distinguished. *Estimation*, in the literature often referred to as *localization*, recovers $\mathbf{T}$ from a single observation without prior pose knowledge, whereas *tracking* updates $\mathbf{T}$ across multiple frames starting from a known initial pose. Because tracking only searches the immediate neighborhood of the previous pose, it is computationally cheaper. In exchange, it depends on an external source of initialization and of recovery after tracking loss. This asymmetry between global search and local update recurs throughout the report and motivates its analysis and central conclusion.

B. Evaluation Metrics

The standard accuracy metric is the average distance of model points, referred to as ADD [14], given the estimated pose $(\hat{\mathbf{R}}, \hat{\mathbf{t}})$, the ground truth $(\mathbf{R}, \mathbf{t})$, and the model point set $\mathcal{M}$, where $\|\cdot\|$ denotes the Euclidean norm:

$$\text{ADD} = \frac{1}{|\mathcal{M}|} \sum_{\mathbf{x} \in \mathcal{M}} \big\| (\hat{\mathbf{R}}\mathbf{x} + \hat{\mathbf{t}}) - (\mathbf{R}\mathbf{x} + \mathbf{t}) \big\|. \tag{3}$$

For objects with geometric symmetries, multiple poses are visually indistinguishable and (3) penalizes correct estimates. A solution is the symmetric variant ADD-S measuring the distance to the closest model point,

$$\text{ADD-S} = \frac{1}{|\mathcal{M}|} \sum_{\mathbf{x}_1 \in \mathcal{M}} \min_{\mathbf{x}_2 \in \mathcal{M}} \big\| (\hat{\mathbf{R}}\mathbf{x}_1 + \hat{\mathbf{t}}) - (\mathbf{R}\mathbf{x}_2 + \mathbf{t}) \big\|. \tag{4}$$

A pose is typically counted as correct [14] if the error remains below 10 % of the object diameter. Results over a sequence are commonly summarized as the area under the accuracy-threshold curve, or AUC. The BOP benchmark [12] additionally reports the average recall, or AR, over three complementary pose-error functions. These comprise the Visible Surface Discrepancy, robust to occlusions and symmetries, the Maximum Symmetry-Aware Surface Distance over model points, and the Maximum Symmetry-Aware Projection Distance in the image plane. The AR is averaged over its core datasets. We use BOP AR whenever comparing methods across papers, since it is the only metric evaluated under a shared protocol. Other scores stem from the respective authors' evaluations and are only comparable with caution.

C. Robotic Deployment Requirements

We evaluate all methods against the following three requirements.

Generalization (R1) The method must localize objects that were not available at training time, given only a CAD model or a few reference images. This requirement is grounded in real-world applications where production lines change parts, households contain open-ended object sets, and per-object retraining is operationally not feasible.

Real-time capability (R2) In closed-loop visual control, the perception latency bounds how quickly the robot can react to changes in the scene and, therefore, its tracking performance during dynamic manipulation [2].

Three rates must be distinguished. The servo loop of a torque-controlled manipulator runs at 100 Hz–1 kHz [2], current cameras deliver frames at 30–60 Hz, and the pose estimator adds processing time on top. Since the measurement rate is bounded by the camera, no camera-based estimator can close the servo loop directly. The control loop therefore always requires a model-based predictor, for example a Kalman filter, that extrapolates the pose between measurements, as realized by Kappler *et al.* [2] through fusing 1 kHz joint encoders with delayed camera observations. Each estimate moreover refers to an image that is one to two frames old through acquisition and transfer, and estimation adds its own delay, which the predictor must also compensate. Pose estimation must therefore feed the predictor at camera frame rate with bounded latency, since the prediction error grows with the extrapolation horizon. As a minimal requirement we propose that the pose is updated at a camera frame rate of at least 30 Hz, corresponding to a per-frame budget of 33 ms.

Accuracy (R3) Grasping tolerances of parallel grippers and industrial insertion tasks are typically in the millimeter range. On public benchmarks, this corresponds to state-of-

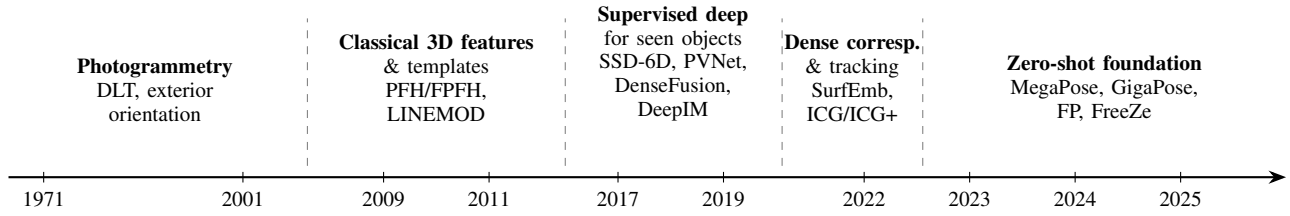


Fig. 1. Development of 6D object pose estimation across the methods analyzed in this report. Each phase relaxed a limitation of its predecessor, moving from known point correspondences, to hand-crafted features on depth data, to learned features for seen objects, to learned correspondence distributions, and finally to novel-object generalization via foundation models. Note that the timeline axis is not to scale.

the-art BOP AR or to the ADD and ADD-S scores rather than merely above-threshold detection.

The requirements refer to the complete perception task, including initialization and recovery. A tracker that sustains real-time rates only after an external system supplies the initial pose does not by itself satisfy R2, because the global search has been delegated, not eliminated.

Second, the thresholds are deliberate choices. The 30 Hz bound for R2 is minimal, and R3 designates the accuracy frontier of the respective benchmark rather than a fixed tolerance, since no published study maps benchmark scores to grasp success rates. The millimeter tolerance motivates R3, the frontier defines it.

III. CLASSICAL APPROACHES

Classical methods address pose estimation with explicit geometry and hand-crafted features rather than learned representations.

A. Geometric Foundations

The earliest solutions predate dense sensing and assume that 2D to 3D point correspondences are given, which reduces pose estimation to linear algebra. The DLT [3], introduced in 1971, solves a homogeneous linear system that maps known object-space coordinates to image coordinates. Because it does not enforce the orthonormality of $\mathbf{R}$, its solution is sensitive to noise. Fiore's exterior-orientation solver [4] removes this weakness by isolating the unknown point depths via null-space projection and then recovering the rotation with a singular value decomposition, with its dominant steps linear in the number of points, while guaranteeing $\mathbf{R} \in SO(3)$. Solvers of this kind are computationally trivial for modern laptops but presuppose solved correspondences, which are not always available in real-world scenarios.

B. Local 3D Descriptors: FPFH

For 3D point clouds the Point Feature Histogram (PFH) [15] proposed a local descriptor that encodes the geometry around a point by histogramming angular relations between the normals of all point pairs in its k-neighborhood, at a cost of $O(n \cdot k^2)$ for n points. Fast Point Feature Histograms (FPFH) [5] reduced this cost to $O(n \cdot k)$ by first computing a simplified histogram $\mathrm{SPFH}(\mathbf{p})$ from each point's own neighbors only, and then re-weighting with the neighbors' histograms according to the normalized weighted sum:

$$\mathrm{FPFH}(\mathbf{p}) = \mathrm{SPFH}(\mathbf{p}) + \frac{1}{k} \sum_{i=1}^{k} \frac{1}{\omega_i} \mathrm{SPFH}(\mathbf{p}_i), \quad (5)$$

where ω_i is the spatial distance between $\mathbf{p}$ and its neighbor $\mathbf{p}_i$, acting as an inverse-distance weight. For registration, the accompanying Sample Consensus Initial Alignment (SAC-IA) repeatedly samples points, retrieves candidates with similar FPFH signatures, and scores the resulting transformations. On the outdoor laser-scan dataset of the original publication, SAC-IA completed alignments in 34 s that the authors' earlier greedy approach had not achieved after 17 minutes, although alignment quality is assessed only qualitatively without a pose-error metric [5]. This latency refers to large-scale scene registration, whereas small tabletop objects, as used in BOP, typically register in a few seconds.

Judged against our requirements of Section II-C, FPFH-based registration occupies an instructive position. It is training-free and object-agnostic, satisfying R1 for rigid geometry, but it is neither real-time at scene scale nor competitive in accuracy with modern learned methods, since its descriptors, built purely on local surface normals, degrade under sensor noise and on symmetric or repetitive geometry. Related classical detectors share this profile. LINEMOD [6], whose follow-up dataset [14] later became a standard benchmark, matches multimodal templates of color gradients and surface normals at about 10 frames per second and detects texture-less objects with true-positive rates of roughly 97–98 % on the authors' six-sequence dataset, but requires capturing templates per object and viewpoint. Point-pair-feature pipelines in the lineage of Drost *et al.* [16], later accelerated by Vock *et al.* [17], provide robust CAD-based detection in industrial point clouds, reaching average recalls of 67.1–94.0 % on four BOP datasets at 4.9 s per scene [17]. In each case, hand-crafted features purchase either generality or speed, but not both together with accuracy.

C. Multi-Modality Tracking: ICG+

An approach to increase speed for classical methods is to avoid global search altogether and instead track the pose across frames. ICG+ [13] extends the authors' earlier depth-and-region tracker ICG with appearance information and represents the current state of the art of classical, optimization-based tracking. It uses four probabilistic modalities, namely

region statistics separating object from background, point-to-plane depth alignment, ORB or SIFT keypoint texture on keyframes, and multi-region information for objects composed of movable parts. Each modality m contributes a likelihood, and the pose is found by maximizing the joint log-likelihood

$$\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} \sum_{m} \ln p(\mathcal{D}_m \mid \boldsymbol{\theta}), \tag{6}$$

optimized with Newton steps under Tikhonov regularization, where $\boldsymbol{\theta} \in \mathbb{R}^6$ parameterizes the pose update and $\mathcal{D}_m$ is the data of modality m.

On the YCB-Video dataset [18], ICG+ attains 93.7 % ADD and 97.7 % ADD-S AUC in its ORB variant, ahead of the deep-learning trackers se(3)-TrackNet with 93.0 %/95.7 % and PoseRBPF with 87.5 %/95.2 %, while running at 312.4 Hz on a single CPU core. The depth-and-region-only baseline ICG reaches 788.4 Hz [13].

Ablations identify depth and region as the essential modalities, since removing them drops ADD to 47.5 % and 22.4 %, respectively, whereas texture and multi-region fusion contribute accuracy refinements.

ICG+ demonstrates that real-time capability is possible together with strong accuracy, for the easier tracking problem. In deployment, its pose update rate remains bounded by the camera frame rate, so the 3.2 ms per-frame cost translates into latency headroom rather than into a higher measurement rate. Nonetheless, the method cannot initialize itself, cannot recover after tracking loss, and requires a textured 3D model of each object. Every capability it demonstrates is therefore conditional on a pose that some other system must first provide.

Classical methods as a whole thus deliver speed and, in tracking, accuracy, yet presuppose either solved correspondences, per-object templates, or an initial pose. Learning-based methods remove the hand-crafted components that impose these preconditions.

IV. LEARNING-BASED APPROACHES

A. Supervised Estimation for Seen Objects

The first deep-learning wave starting 2017 replaced hand-crafted features with convolutional networks trained on a fixed set of objects. SSD-6D [19] casts rotation estimation as viewpoint classification over renderings of a CAD model, and Deep-6DPose [20] regresses poses directly within a Mask R-CNN-style detector. PVNet [7] lets every object pixel vote for the directions toward a small set of keypoints and solves an uncertainty-weighted PnP problem, reaching 86.3 % ADD on the LINEMOD dataset [14] at roughly 25 frames per second. DenseFusion [8] fuses per-pixel CNN color features with PointNet geometry features and predicts a confidence-weighted pose per pixel, reaching 93.1 % ADD-S AUC on YCB-Video [18] at about 16 frames per second.

DeepIM [21] introduced learned render-and-compare refinement. It iteratively renders the current pose estimate and predicts a disentangled SE(3) update from the discrepancy between rendering and observation, lifting the ADD score of PoseCNN on LINEMOD from 62.7 % to 88.6 % at roughly 12 frames per second per object. This refinement paradigm reappears in nearly all later methods. The voting idea was also transferred to point clouds [22], with G2L-Net [23] reaching 98.7 % ADD on LINEMOD at a real-time rate of 23 frames per second.

Collectively, these methods established state-of-the-art accuracy on public benchmarks, several of them at latencies between 40 and 100 ms, near but above the 33 ms budget of Section II-C.

Their common limitation is generalization. Each network is trained for a fixed object set, and every additional object costs hours of synthetic data generation and retraining.

B. Dense Correspondence Distributions: SurfEmb

SurfEmb [9] carried the correspondence idea to its most expressive form within the seen-object paradigm. Instead of a sparse set of keypoints, it learns continuous distributions over dense 2D to 3D correspondences. An image encoder Φ and a surface-coordinate encoder Ψ are trained with a contrastive InfoNCE loss such that, for each object pixel $\mathbf{u}$, the probability of surface point $\mathbf{x}$ is

$$p(\mathbf{x} \mid \mathbf{u}) = \frac{\exp\big(\Phi(\mathbf{u})^\top \Psi(\mathbf{x})\big)}{\sum_{\mathbf{x}' \in \mathcal{M}} \exp\big(\Phi(\mathbf{u})^\top \Psi(\mathbf{x}')\big)}. \tag{7}$$

Crucially, symmetries require no manual annotation. For ambiguous views, (7) is multi-modal and assigns comparable probability to all symmetry-equivalent surface points. Poses are generated with a PnP solver on sampled correspondences, scored against the predicted distributions, and refined by maximizing correspondence likelihood. Trained purely on synthetic images, SurfEmb reached the state of the art of its time on the BOP core datasets with a mean AR of 75.2 % with depth refinement [9]. The method remains per-object trained, and inference takes roughly 2.2 s per detection, dominated by PnP-RANSAC and refinement rather than by the network itself.

C. Zero-Shot Estimation with Foundation Models

The zero-shot paradigm trains a model once on millions of synthetic scenes rendered from diverse CAD models, or forgoes training altogether, and localizes any novel object at test time.

MegaPose [24] extended the DeepIM-style render-and-compare approach to novel objects by training a coarse classifier and a refiner on two million synthetic images of diverse objects.

It reaches 57.2 % mean BOP AR over the seven core datasets, despite never having seen the test objects. Its coarse stage compares hundreds of renderings per detection and requires $\sim$2.5 s in the paper's own measurement while a single refiner iteration costs $\sim$50 ms.

This bottleneck is addressed by GigaPose [25]. It retrieves the out-of-plane rotation from only 162 pre-embedded templates using DINOv2 [26] patch features and regresses the remaining four degrees of freedom from a single 2D to 2D patch correspondence, reducing coarse estimation to 48 ms, a

35× speedup over the 1.68 s the authors measure for MegaPose, while being markedly more robust to segmentation noise.

FoundationPose (FP) [10] unifies model-based and model-free operation in one system. If no CAD model exists, it builds a neural implicit object representation, in the form of a signed distance field with an appearance field from roughly 16 reference images. Either representation then feeds a render-and-compare pipeline in which a transformer-based refiner iteratively updates sampled pose hypotheses,

$$\mathbf{T}_{k+1} = \Delta\mathbf{T}_k \mathbf{T}_k, \tag{8}$$

$$\Delta\mathbf{T}_k = f_{\text{ref}}\big(I_{\text{obs}},\, I_{\text{rend}}(\mathbf{T}_k)\big), \tag{9}$$

where $\mathbf{T}_k$ is the pose hypothesis after k refinement iterations and the refiner f_{ref} predicts from the observed crop I_{obs} and the rendering $I_{\text{rend}}(\mathbf{T}_k)$ the update $\Delta\mathbf{T}_k$, parameterized in the camera frame, which justifies the left-multiplication in (8). A hierarchical ranking network then scores all refined hypotheses jointly, attending across hypotheses rather than scoring each in isolation, to select the final pose.

Trained exclusively on synthetic data whose textures are diversified with LLM-guided diffusion augmentation, FP reaches 83.3 % mean BOP AR on LM-O, T-LESS, and YCB-V, ahead of MegaPose on the same three datasets (58.6 %) and even ahead of the instance-level SurfEmb+ICP (79.7 %) [10]. On the continuously updated BOP-Classic-Core leaderboard, which evaluates per image over all seven core datasets, FP reaches 73.4 % AR at 29.3 s per image [12].

Single-frame estimation costs roughly 1.3 s, dominated by refinement and hypothesis ranking, but once initialized, the refiner alone tracks at roughly 32 Hz. A single network spans two operating points separated by a factor of forty in latency, depending only on the scope of the pose search.

FreeZe [11] demonstrates that zero-shot pose estimation can be achieved without any training. It fuses geometric descriptors computed on the CAD point cloud with DINOv2 visual features back-projected onto the depth point cloud, obtains 3D to 3D correspondences between query and model, and solves for the pose with RANSAC and ICP, followed by a symmetry-aware refinement that prunes symmetry-equivalent hypotheses using visual similarity.

With zero task-specific training, FreeZe reaches 69.3 % mean BOP AR with refinement [11]. In the BOP Challenge 2024 [12], its improved variant FreeZeV2.1 won model-based 6D localization of unseen objects with 82.1 % AR, only 3.5 percentage points behind the best method for seen objects, GPose2023 with 85.6 % AR, which has no accompanying publication yet. On the leaderboard, its current variant FreeZeV2.2 reaches 83.3 % AR at 21.7 s per image.

The resulting 9.9-point margin of FreeZeV2.2 over FP is concentrated where appearance is least informative. On the largely textured objects of LM-O, TUD-L, HB, and YCB-V, the two methods lie within two to six points, but FreeZe leads by 15.3 points on the texture-less, symmetric industrial parts of T-LESS, by 17.4 points on ITODD, and by 20.3 points on the heavily occluded bin-picking scenes of IC-BIN.

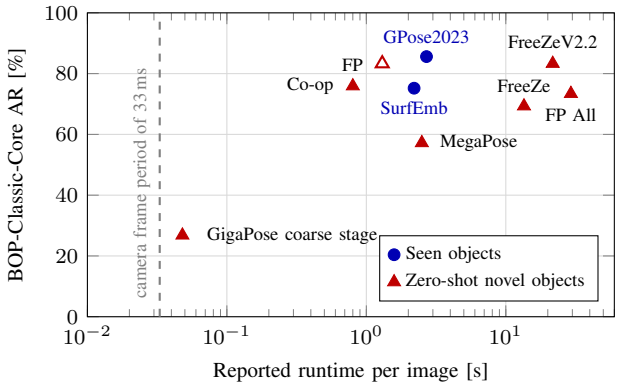


Fig. 2. Accuracy versus runtime for single-frame pose estimation on the BOP-Classic-Core datasets. Runtimes are as reported by the respective authors or by the BOP evaluation [12] on differing hardware. Circles denote object-specifically trained methods evaluated on seen objects. FP appears twice: the open triangle denotes its published AR averaged over LM-O, T-LESS, and YCB-V only, the filled triangle marked "FP All" its per-image leaderboard result over all seven core datasets. GigaPose is represented by its coarse stage only.

FreeZe anchors the pose primarily in pretrained geometric descriptors matched between the CAD model and the depth point cloud, with the visual features chiefly resolving symmetry and texture ambiguity, so its local 3D correspondences with RANSAC and ICP remain informative on texture-less surfaces and under partial occlusion. FP instead scores rendered hypotheses against the observed crop with networks trained on synthetic scenes, an appearance-driven comparison whose published 83.3 % averages LM-O, T-LESS, and YCB-V only and falls to 73.4 % once the industrial and bin-picking datasets enter the average. Part of that drop originates upstream. The publication's T-LESS score of 83.0 % relies on dataset-specific Mask R-CNN detections, whereas leaderboard submissions must localize unseen objects with zero-shot detectors, the bottleneck Section V returns to.

For unseen objects, training-free foundation-feature fusion has therefore essentially caught up with purpose-trained, object-specific pipelines. The only remaining cost is runtime. FreeZeV2.1 requires 24.9 s per image against 2.7 s for GPose2023. The BOP 2024 report further documents Co-op as a more practical operating point, reaching 75.9 % AR at 0.8 s per image, and identifies the zero-shot 2D detection stage as a principal remaining accuracy bottleneck [12].

Real-time capability remains unmet. No zero-shot estimator localizes objects at camera frame rate, and the fastest complete pipelines operate at roughly one second per image, against the 3.2 ms at which ICG+ tracks.

V. DISCUSSION

A. *Trade-off Analysis*

Table I consolidates the evidence of the two preceding sections, and Fig. 2 visualizes the estimation methods on the shared BOP metric. Classical registration and template matching such as FPFH and LINEMOD are object-agnostic but neither sufficiently accurate nor, at scene scale, sufficiently fast. Supervised networks like PVNet, DenseFusion

TABLE I

REPRESENTATIVE METHODS EVALUATED AGAINST THE REQUIREMENTS R1–R3, DETAILED IN SECTION II-C. RUNTIMES AND SCORES ARE AS REPORTED IN THE CITED PAPERS OR BY BOP 2024 ON DIFFERING HARDWARE. LATENCIES ARE GIVEN IN MILLISECONDS PER FRAME, WHERE VALUES MARKED $\sim$ ARE CONVERTED FROM THE FRAME RATES REPORTED IN THE CITED PAPERS. "EST." DENOTES SINGLE-FRAME ESTIMATION AND "TR." DENOTES FRAME-TO-FRAME TRACKING.

Method	Input	Mode	Novel objects	Latency [ms]	Reported accuracy	Unmet
FPFH + SAC-IA [5]	Depth	Est.	✓[a]	34 000[e]	n/a[a]	R2, R3
LINEMOD [6]	RGB-D	Est.	×	∼100	97–98 % detection[b]	R1, R2
PVNet [7]	RGB	Est.	×	∼40	86.3 % ADD (LINEMOD)	R1, R2
DenseFusion [8]	RGB-D	Est.	×	∼63	93.1 % ADD-S AUC (YCB-V)	R1, R2
DeepIM [21]	RGB	Est.	×	∼83	88.6 % ADD (LINEMOD)	R1, R2
SurfEmb [9]	RGB or RGB-D	Est.	×	∼2 200	75.2 % BOP AR	R1, R2
ICG+ [13]	RGB-D	Tr.	✓[f]	3.2	97.7 % ADD-S AUC (YCB-V)	R2[c]
MegaPose [24]	RGB or RGB-D	Est.	✓	2 500	57.2 % BOP AR	R2
GigaPose [25]	RGB	Est.	✓	48	26.8 % BOP AR (coarse)	R2, R3
FoundationPose [10], [12]	RGB-D	Est. / Tr.	✓	29 300[d] / ∼31	73.4 % BOP AR[d] / 97.9 % ADD-S AUC (YCB-V)	R2 (Est.)
FreeZeV2.1 [12]	RGB-D	Est.	✓	24 900	82.1 % BOP AR	R2
Co-op [12]	RGB-D	Est.	✓	800	75.9 % BOP AR	R2

[a] Object-agnostic for rigid geometry, but registration-based and not benchmarked on BOP-style localization. [b] Detection rate on the authors' own dataset, not a pose-error metric. [c] Additionally requires external pose initialization and recovery as detailed in Section III-C. [d] Per-image result from the BOP leaderboard over all seven core datasets; the paper reports 1.3 s per object. [e] Per alignment on the outdoor laser-scan dataset of the original publication. [f] Object-agnostic at test time given a 3D model of the novel object; the keypoint modality additionally requires surface texture.

and SurfEmb achieve accuracy at latencies close to, but above, the 33 ms budget, and only for the objects they were trained on. Zero-shot foundation methods such as MegaPose, FP, and FreeZe achieve generalization and accuracy at runtimes of seconds to tens of seconds. GigaPose's coarse stage comes closest to the budget at 48 ms, but still exceeds it and trades away the accuracy frontier. The one method operating inside the real-time regime, ICG+, addresses the strictly easier tracking problem and presupposes an initialized pose and a per-object model.

Every method in Table I violates at least one of the three requirements. The strongest candidate for a counterexample is FP in tracking mode, which follows novel objects at 32 Hz with state-of-the-art accuracy. Under the definitions of Section II, however, tracking presupposes an initialized pose. Whenever the object must first be acquired, or re-acquired after occlusion, the 1.3 s estimation mode re-enters the loop. Section V-B returns to this two-mode structure.

The hypothesis therefore stands, although its precise boundary depends on the thresholds stated in Section II-C. The binding constraint has shifted from accuracy in the classical era, to generalization in the supervised era, to latency in the foundation-model era.

B. Decoupled Hybrid Architectures

If no single method satisfies all three requirements, the architectural conclusion is to stop demanding this of a single method. Under the strict reading of R2 in Section II-C, which charges initialization and recovery to the perception system, a hybrid does not eliminate the global search either. It bounds the time during which no pose is available. We therefore evaluate hybrid architectures against a relaxed criterion, namely that the steady-state pose rate meets the 30 Hz bound while initialization and recovery complete within a bounded, application-defined latency. A monolithic zero-shot estimator fails even this relaxed criterion, and a tracker alone still fails it because, without an initializer, its recovery latency is unbounded.

Decoupling initialization from tracking is not a hypothetical construction. FP already implements it within a single network. The identical refiner that needs 1.3 s for global estimation tracks at 32 Hz once the search collapses to the neighborhood of the previous pose [10]. MegaPose exhibits the same structure, with seconds for the coarse search, roughly 50 ms per refinement iteration, and 20 Hz tracking [24]. In every zero-shot estimator, the expensive operation is the global search over the pose space of a novel object, while the per-frame update is cheap.

FP's 32 Hz tracking sits directly at the 30 Hz bound of R2. Its per-frame cost consumes nearly the entire frame period, adding almost a full frame of latency and leaving no headroom for catch-up after re-initialization. Moreover, both of its operating modes occupy the same GPU, so a 1.3 s re-initialization after tracking loss starves the tracker of compute at exactly the moment the object moves unpredictably. A heterogeneous decoupling avoids both limits. ICG+ processes a frame in 3.2 ms on a single CPU core. Its update rate remains bounded by the camera, but the tenfold headroom below the 33 ms frame period keeps the added latency negligible and accelerates the catch-up replay, while the GPU remains free to run the zero-shot estimator asynchronously for initialization and recovery. The contribution of a hybrid is thus not the decoupling itself but its distribution across heterogeneous compute. ICG+'s reported accuracy stems from seen, textured YCB-Video objects, however, and its transfer to arbitrary novel objects is unvalidated, with the depth-and-region variant ICG as the fallback for texture-less parts.

This suggests the pipeline sketched in Fig. 3. GigaPose's 48 ms coarse stage followed by a refiner, or FP's estimation mode are zero-shot initializers, which run once at startup

and asynchronously afterwards for recovery, while a classical multi-modality tracker such as ICG+ supplies pose measurements at the full camera frame rate in between. A model-based predictor then extrapolates these measurements to the servo rate of the controller, as required by Section II-C. To resolve the latency of the asynchronous initializer, which runs on a historical frame and produces a stale pose by the time computation finishes, the pipeline incorporates a latency-compensation step, maintaining a sliding history buffer of camera frames and tracking states. Specifically, let I_τ be the RGB-D frame with frame index τ, and let $\Delta_f = \lceil \Delta f_{\mathrm{cam}} \rceil$ be the initialization latency Δ expressed in frames at the camera rate f_{cam}. Once the initializer returns a pose estimate $\mathbf{T}^{\mathrm{init}}_{t-\Delta_f}$ for the past frame $t - \Delta_f$, we synchronize the state by sequentially running the local tracker f_{track} on the intermediate buffered frames:

$$\mathbf{T}_{t-\Delta_f+k} = f_{\mathrm{track}}\big(\mathbf{T}_{t-\Delta_f+k-1},\, I_{t-\Delta_f+k}\big) \qquad (10)$$

for $k = 1, \ldots, \Delta_f$, starting with $\mathbf{T}_{t-\Delta_f} = \mathbf{T}^{\mathrm{init}}_{t-\Delta_f}$. This effectively recovers the pose estimate to the current frame t before resuming high-rate tracking and control. Because new frames continue to arrive during the replay, the catch-up terminates only if the tracker's per-frame cost remains well below the camera frame period. ICG+ at 3.2 ms against a 33 ms period replays roughly ten buffered frames per newly arriving frame, and the required buffer depth is bounded by the worst-case initializer latency, about 40 frames for FP's 1.3 s.

The individual components are validated in existing literature. Haugaard and Buch [9] position dense estimation as initialization for trackers, Stoiber *et al.* [13] state the need for external initialization, and the BOP 2024 report states that further speed-ups are needed for real-time applications [12]. We present the combined pipeline as a development perspective, whose plausibility rests on the published component latencies, not on end-to-end experiments.

Recovery after tracking loss is the critical failure mode of this architecture. During the latency window following a loss trigger, no validated pose is available, and the control layer must bridge this blind window explicitly, for example by holding the last pose, propagating it with a motion model, or retreating to a safe state. Moreover, the catch-up replay reuses exactly the buffered frames on which tracking previously failed, so a persisting disturbance such as occlusion can fail the replay again and trap the system in repeated re-initialization. The recovery logic must detect this case and instead re-initialize on the newest available frame.

The pose filter between tracker and controller is a structural component of this architecture rather than an optional refinement. Even in steady state, the tracker output refers to an image that is one to two frames old, carries the tracker's processing delay, and arrives below the servo rate of Section II-C. An error-state Kalman filter formulated on $SE(3)$ serves three functions. First, it extrapolates the pose to the 100 Hz–1 kHz ticks of the servo loop and compensates the constant acquisition and processing latency. Second, rather

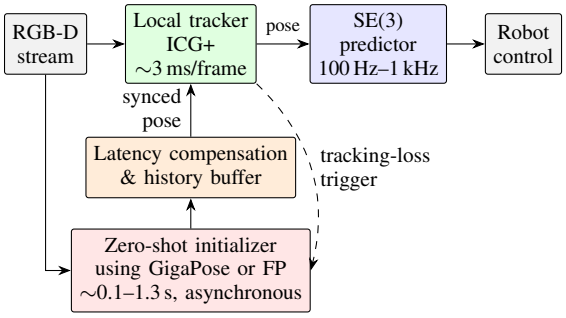


Fig. 3. Decoupled hybrid architecture as a development perspective. A zero-shot estimator provides novel-object initialization and asynchronous recovery (R1, R3), whose output is synchronized via a latency-compensation history buffer with the classical tracker maintaining the pose at camera frame rate (R2). An SE(3) predictor extrapolates the camera-rate measurements to the servo rate of the robot controller and compensates the acquisition and processing latency.

than applying a hard reset with $\mathbf{T}^{\mathrm{init}}_{t-\Delta_f}$, it merges the initializer's output with the tracking history, thereby smoothing transitions and mitigating local tracking drift accumulated during the latency window. Third, it gates outliers, because an initializer estimate that lands on a symmetry-equivalent mode would otherwise corrupt a healthy track. It remains an open engineering question how tracking loss is detected reliably and how the architecture scales to scenes containing many objects.

C. Open Problems and Development Perspectives

Beyond the engineering of hybrid pipelines, three problems remain open even under a hybrid architecture. The first is the 2D detection bottleneck. Zero-shot pose estimators consume 2D detections or segmentation masks of unseen objects, and the BOP 2024 report identifies this upstream stage as a principal accuracy limit [12]. When the detector misses a heavily occluded object, no downstream stage can recover it. The second is object-model dependence. Model-based operation assumes an accurate CAD model, and model-free operation still requires on the order of 16 reference images [10]. Single-reference approaches exist but trail model-based accuracy [27]. For truly open-world manipulation, object onboarding must become as cheap as pointing a camera. In addition, severe occlusions, lack of surface texture, and optical properties like specularity or transparency remain open challenges, since foundation features such as those from DINOv2 rely on semantic textures and struggle on reflective, transparent, or visually indistinguishable objects. The third is rigidity. Every method surveyed here assumes a rigid object whose pose is an element of $SE(3)$. Articulated and deformable objects, such as cables, packaging, or textiles, fall outside the problem formulation itself, not merely outside current methods' capabilities.

Benchmark realism compounds these concerns. The egocentric scenes of BOP-H3 reduce the accuracy of pipelines that excel on the classic datasets by more than a third [12], and the runtimes in Table I stem from heterogeneous hardware. A shared latency-aware benchmark, extending the

runtime reporting that the BOP organizers already provide, would resolve the latter ambiguity and measure pose estimation under the constraint that matters most for closed-loop robotics.

VI. CONCLUSIONS

This report analyzed the last five decades of 6D object pose estimation through the lens of the three robotic deployment requirements of novel-object generalization, real-time capability, and accuracy. The published evidence supports the clear conclusion that no single current method satisfies all three. The supervised deep-learning era bought accuracy at the price of per-object training, a price the foundation-model era has largely eliminated, as a training-free pipeline came within 3.5 percentage points AR of the best seen-object method in the BOP Challenge 2024. It pays instead with runtimes one to three orders of magnitude above the 33 ms frame budget of camera-rate perception, while the fastest tracker in the literature processes a frame in 3.2 ms yet cannot initialize itself. The published runtime profiles consistently locate the expense in the global pose search rather than in the per-frame update, a decoupling that FP already exploits between its 1.3 s estimation mode and its 32 Hz tracking mode. Building on this principle, heterogeneous hybrid architectures, which place zero-shot initialization and recovery around a classical camera-rate tracker and bridge the remaining gap to the servo rate with a model-based predictor, emerge as the most viable near-term path to satisfying all three requirements jointly. Demonstrating such a pipeline end-to-end under a latency-aware evaluation protocol extending BOP's runtime reporting may be a next step for the field.

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REFERENCES

[1] C. Yuanwei, M. Hairi Mohd Zaman, and M. Faisal Ibrahim, "A review on six degrees of freedom (6D) pose estimation for robotic applications," *IEEE Access*, vol. 12, pp. 161 002–161 017, 2024.

[2] D. Kappler *et al.*, "Real-time perception meets reactive motion generation," *IEEE Robotics Autom. Lett.*, vol. 3, no. 3, pp. 1864–1871, Jul. 2018.

[3] Y. Abdel-Aziz and H. Karara, "Direct linear transformation from comparator coordinates into object space coordinates in close-range photogrammetry," *Photogramm. Eng. Remote Sens.*, vol. 81, no. 2, pp. 103–107, Feb. 2015, reprint of the 1971 original.

[4] P. Fiore, "Efficient linear solution of exterior orientation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 23, no. 2, pp. 140–148, Feb. 2001.

[5] R. B. Rusu, N. Blodow, and M. Beetz, "Fast point feature histograms (FPFH) for 3D registration," in *Proc. IEEE Int. Conf. Robotics Autom. (ICRA)*. Kobe: IEEE, May 2009, pp. 3212–3217.

[6] S. Hinterstoisser *et al.*, "Multimodal templates for real-time detection of texture-less objects in heavily cluttered scenes," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*. Barcelona, Spain: IEEE, Nov. 2011, pp. 858–865.

[7] S. Peng, Y. Liu, Q. Huang, X. Zhou, and H. Bao, "PVNet: Pixel-wise voting network for 6DoF pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*. Long Beach, CA, USA: IEEE, Jun. 2019, pp. 4556–4565.

[8] C. Wang *et al.*, "DenseFusion: 6D object pose estimation by iterative dense fusion," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*. Long Beach, CA, USA: IEEE, Jun. 2019, pp. 3338–3347.

[9] R. L. Haugaard and A. G. Buch, "SurfEmb: Dense and continuous correspondence distributions for object pose estimation with learnt surface embeddings," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*. New Orleans, LA, USA: IEEE, Jun. 2022, pp. 6739–6748.

[10] B. Wen, W. Yang, J. Kautz, and S. Birchfield, "FoundationPose: Unified 6D pose estimation and tracking of novel objects," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*. Seattle, WA, USA: IEEE, Jun. 2024, pp. 17 868–17 879.

[11] A. Caraffa, D. Boscaini, A. Hamza, and F. Poiesi, "FreeZe: Training-free zero-shot 6D pose estimation with geometric and vision foundation models," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, A. Leonardis, E. Ricci, S. Roth, O. Russakovsky, T. Sattler, and G. Varol, Eds. Cham: Springer Nature Switzerland, 2025, vol. 15133, pp. 414–431.

[12] V. N. Nguyen *et al.*, "BOP Challenge 2024 on model-based and model-free 6D object pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR) Workshops*, 2025.

[13] M. Stoiber, M. Elsayed, A. E. Reichert, F. Steidle, D. Lee, and R. Triebel, "Fusing visual appearance and geometry for multi-modality 6DoF object tracking," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*. Detroit, MI, USA: IEEE, Oct. 2023, pp. 1170–1177.

[14] S. Hinterstoisser *et al.*, "Model based training, detection and pose estimation of texture-less 3D objects in heavily cluttered scenes," in *Proc. Asian Conf. Comput. Vis. (ACCV)*, D. Hutchison *et al.*, Eds. Berlin, Heidelberg: Springer Berlin Heidelberg, 2013, vol. 7724, pp. 548–562.

[15] R. Rusu, N. Blodow, Z. Marton, and M. Beetz, "Aligning point cloud views using persistent feature histograms," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*. Nice: IEEE, Sep. 2008, pp. 3384–3391.

[16] B. Drost, M. Ulrich, N. Navab, and S. Ilic, "Model globally, match locally: Efficient and robust 3D object recognition," in *Proc. IEEE Comput. Soc. Conf. Comput. Vis. Pattern Recognit.* San Francisco, CA, USA: IEEE, Jun. 2010, pp. 998–1005.

[17] R. Vock, A. Dieckmann, S. Ochmann, and R. Klein, "Fast template matching and pose estimation in 3D point clouds," *Computers & Graphics*, vol. 79, pp. 36–45, Apr. 2019.

[18] Y. Xiang, T. Schmidt, V. Narayanan, and D. Fox, "PoseCNN: A convolutional neural network for 6D object pose estimation in cluttered scenes," in *Proc. Robotics: Science and Systems*, Pittsburgh, Pennsylvania, Jun. 2018.

[19] W. Kehl, F. Manhardt, F. Tombari, S. Ilic, and N. Navab, "SSD-6D: Making RGB-based 3D detection and 6D pose estimation great again," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*. Venice: IEEE, Oct. 2017, pp. 1530–1538.

[20] T.-T. Do, M. Cai, T. Pham, and I. Reid, "Deep-6DPose: Recovering 6D object pose from a single RGB image," arXiv preprint arXiv:1802.10367, 2018.

[21] Y. Li, G. Wang, X. Ji, Y. Xiang, and D. Fox, "DeepIM: Deep iterative matching for 6D pose estimation," *Int. J. Comput. Vis.*, vol. 128, no. 3, pp. 657–678, Mar. 2020.

[22] W. Chen, J. Duan, H. Basevi, H. J. Chang, and A. Leonardis, "PointPoseNet: Point pose network for robust 6D object pose estimation," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2020.

[23] W. Chen, X. Jia, H. J. Chang, J. Duan, and A. Leonardis, "G2L-Net: Global to local network for real-time 6D pose estimation with embedding vector features," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*. Seattle, WA, USA: IEEE, Jun. 2020, pp. 4232–4241.

[24] Y. Labbé *et al.*, "MegaPose: 6D pose estimation of novel objects via render & compare," in *Proc. 6th Conf. Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, K. Liu, D. Kulic, and J. Ichnowski, Eds., vol. 205. PMLR, Dec. 2023, pp. 715–725.

[25] V. N. Nguyen, T. Groueix, M. Salzmann, and V. Lepetit, "GigaPose: Fast and robust novel object pose estimation via one correspondence," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*. Seattle, WA, USA: IEEE, Jun. 2024, pp. 9903–9913.

[26] M. Oquab *et al.*, "DINOv2: Learning robust visual features without supervision," *Trans. Mach. Learn. Res.*, 2024.

[27] M. Liu, S. Li, A. Chhatkuli, P. Truong, L. V. Gool, and F. Tombari, "One2Any: One-reference 6D pose estimation for any object," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*. Nashville, TN, USA: IEEE, Jun. 2025, pp. 6457–6467.